

Project: Poker Game Simulator with Expectiminimax

Name:MD.Sabbir Hossain

ID:2212642042

Course:CSE440

Section :1

Group: 7

Week 1 (10 Feb – 16 Feb)

I read the course manual carefully to understand the project milestones and reporting expectations. During group discussions I clarified that my role would focus mainly on literature review, scenario design, and explaining the probabilistic reasoning involved in the AI algorithm. I also reviewed the basic rules of Texas Hold'em poker.

Week 2 (17 Feb – 23 Feb)

I searched for related poker AI implementations and research discussions to understand how others manage large game trees and uncertain information. I summarized ideas such as simplified betting models, depth limits, and state abstraction, and shared these notes with the group.

Week 3 (24 Feb – 1 Mar)

I designed a number of scenario-based tests for the early simulator, including situations where players fold early, where multiple players reach showdown, and where strong shared boards appear. I ran these scenarios using early versions of the simulator and observed whether the outcomes were consistent with expected poker rules.

Week 4 (2 Mar – 8 Mar)

I focused on understanding how chance nodes will be handled in the Expectiminimax implementation. I discussed with the team how unknown cards might be modeled and checked that card draws follow the logic of a standard 52-card deck without duplication.

Week 5 (9 Mar – 15 Mar)

As the AI logic started being integrated into the simulator, I began preparing explanatory materials that illustrate how MAX, MIN, and CHANCE nodes appear during a poker hand. I also collected example scenarios that will help demonstrate the AI decision process later in presentations.

Current Status and Next Steps

My contributions so far include related-work research, scenario-based testing, and preparation of explanatory materials for the AI model. In the coming weeks I plan to refine these explanations, assist with preparing diagrams for the final report and presentation, and help document how the algorithm operates within the simulator.